

Name: _____ COE-email: _____

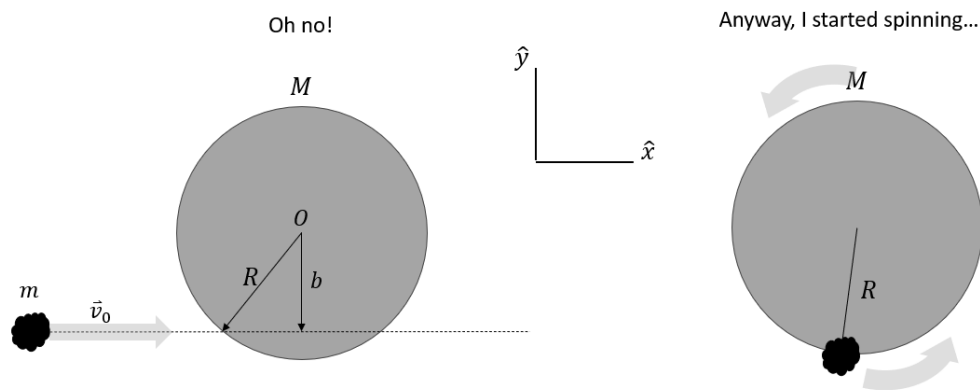
You will receive 1 credit upon complete **all** the tasks.

Each successful task will get you one token for late submission or assignment waiver. Please refer to the course syllabus for token usage.

Task #1: Angular Momentum Conservation in an Inelastic Impact

A uniform solid disk of mass M and radius R lies in a horizontal plane and is mounted on a frictionless vertical spindle passing through its center O . The disk is initially at rest.

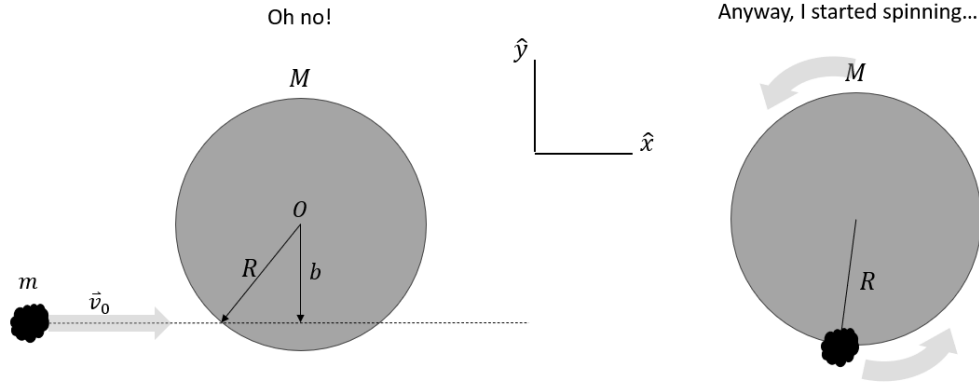
A small sticky blob of putty of mass m slides along the horizontal table toward the disk at a constant velocity $\vec{v}_0$. The line of motion of the putty lies at an impact parameter b ($0 < b < R$) relative to the disk's center O . The putty strikes the disk and sticks to its outer edge at distance R from the center O .



- (a) [1 token] Write down the expression for the initial total angular momentum $\vec{L}_i$ of the system about the center O immediately before the impact. State its magnitude and direction.

$$\vec{L}_i = \vec{r} \times \vec{p} \implies L_i = rp \sin \theta = mv_0 b$$

Direction is: $\odot$



(b) [1 token] Derive the total moment of inertia I_{total} of the system after the collision. Using angular momentum conservation, derive an expression for the final angular velocity ω_f of the disk-putty system in terms of m , M , $\vec{v}_0$, b and R . Hint: moment of inertia of uniform disk is: $I_{\text{disk}} = \frac{1}{2}MR^2$.

$$I_{\text{tot}} = I_{\text{disk}} + I_{\text{putty}} = \frac{1}{2}MR^2 + mR^2 = \left(\frac{M}{2} + m\right) R^2$$

Conservation of angular momentum:

$$L_i = L_f \implies mv_0b = I_{\text{tot}}\omega_f$$

$$\omega_f = \frac{mv_0b}{\left(\frac{M}{2} + m\right) R^2} = \left(\frac{m}{m + \frac{M}{2}}\right) \frac{v_0b}{R^2}$$

(c) [1 token] Calculate the ratio of the final mechanical energy E_f to the initial mechanical energy E_i of the system.

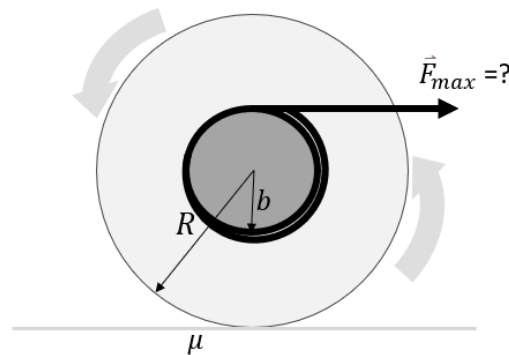
$$E_f = \frac{1}{2}I_{\text{tot}}\omega_f^2 = \frac{1}{2} \left[\left(\frac{M}{2} + m\right) R^2 \right] \left[\frac{mv_0b}{\left(\frac{M}{2} + m\right) R^2} \right]^2 = \frac{m^2v_0^2b^2}{2\left(\frac{M}{2} + m\right) R^2}$$

$$\frac{E_f}{E_i} = \frac{\frac{m^2v_0^2b^2}{(M+2m)R^2}}{\frac{1}{2}mv_0^2} = \frac{2mb^2}{(M+2m)R^2} = \left(\frac{m}{m + \frac{M}{2}}\right) \left(\frac{b}{R}\right)^2$$

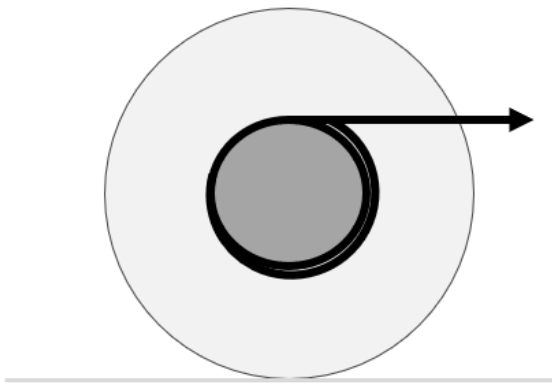
Task #2: Dynamics of a Yoyo Pulled Along the Ground

A yoyo consists of two outer disks of radius R connected by a narrow central axle of radius b ($b < R$). The total mass of the yoyo is m , and its moment of inertia about the center of mass (CM) is given by $I_{cm} = \gamma m R^2$, where γ is a dimensionless shape factor ($0 < \gamma < 1$).

The yoyo rests on a rough horizontal floor with a coefficient of static friction μ . A string is wrapped around the top of the inner axle (at height $R + b$ above the ground) and is pulled horizontally to the right with a constant pulling force $\vec{F}$. The yoyo rolls without slipping.



- (a) [1 token] Draw a free-body diagram below of the yoyo. Set up Newton's second law for linear translational motion and rotational motion about the center of mass.



For translation:

$$F - f_s = ma$$

For rotation:

$$\sum \tau_{\text{cm}} = f_s R + Fb = I_{\text{cm}} \alpha = (\gamma m R^2) \frac{a}{R} = \gamma m R a$$

(b) [1 token] Derive the expression for the static friction force f_s acting on the yoyo in terms of $\vec{F}$, γ , R , μ and b . Under what specific condition (in terms of γ , R and b) will the static friction force be exactly zero ($f_s = 0$)?

$$f_s R + Fb = \gamma m R \left(\frac{F - f_s}{m} \right) = \gamma R (F - f_s)$$

$$f_s R + f_s (\gamma R) = \gamma R F - Fb$$

$$f_s R (1 + \gamma) = F(\gamma R - b)$$

$$f_s = F \frac{\gamma R - b}{R(1 + \gamma)}$$

When $f_s = 0$:

$$\gamma R - b = 0 \implies b = \gamma R \quad \text{or} \quad \gamma = \frac{b}{R}$$

(c) [1 token] Find the maximum force $\vec{F}_{\text{max}}$ that can be applied to the string without causing the yoyo to slip on the floor.

Non-slipping:

$$|f_s| \leq (f_s)_{\text{max}} = \mu_s N = \mu_s mg.$$

$$\left| F \frac{\gamma R - b}{R(1 + \gamma)} \right| \leq \mu_s mg$$

$$F_{\text{max}} = \mu_s mg \frac{R(1 + \gamma)}{|\gamma R - b|}$$